

Curriculum Vitae: Anton Rodomanov

PERSONAL INFORMATION

- Born on 22/01/1994, Russian citizenship, married, 1 daughter.
- E-mail: anton.rodomanov@cispa.de.
- Web-page: arodomanov.github.io.
- Address: Saarbrücken, Germany.
- Languages: English (fluent), German (intermediate), French (intermediate), Russian (native).

RESEARCH INTERESTS

Convex Optimization, Numerical Algorithms, Complexity Estimates, Randomized Methods, Machine Learning, Statistics.

WORK EXPERIENCE

Postdoctoral Researcher CISPA Helmholtz Center for Information Security .	01/09/2023 – now Saarbrücken, Germany
Postdoctoral Researcher ICTEAM Institute at UCLouvain .	01/09/2022 – 31/08/2023 Louvain-la-Neuve, Belgium
Doctoral Candidate Department of Mathematical Engineering (INMA) at UCLouvain .	23/01/2019 – 31/08/2022 Louvain-la-Neuve, Belgium
Lecturer Samsung-HSE Lab at Higher School of Economics .	02/10/2017 – 31/08/2018 Moscow, Russia
Research Assistant International Laboratory of Deep Learning and Bayesian Methods at Higher School of Economics .	09/01/2017 – 18/01/2019 Moscow, Russia

EDUCATION

PhD in Mathematical Engineering Catholic University of Louvain (UCLouvain) , Department of Mathematical Engineering (INMA) Thesis: Quasi-Newton Methods with Provable Efficiency Guarantees . Advisor: Yurii Nesterov .	2019–22 Louvain-la-Neuve, Belgium
MSc in Computer Science (GPA: 9.4/10) Higher School of Economics , Faculty of Computer Science Thesis: A Superlinearly-Convergent Proximal Newton-Type Method for the Optimization of Finite Sums. Advisors: Dmitry Kropotov and Dmitry Vetrov .	2015–17 Moscow, Russia
BSc in Computer Science (GPA: 4.8/5) Lomonosov Moscow State University , Faculty of Computational Mathematics and Cybernetics Thesis: Development of a Stochastic Optimization Method for Machine Learning Problems with Big Data. Advisors: Dmitry Kropotov and Dmitry Vetrov .	2011–15 Moscow, Russia

PUBLICATIONS

Preprints

Efficient Gradient Methods for Distributed Saddle Problems R. Luo, A. Rodomanov, S. Stich. [arXiv]	2026
Global Complexity Analysis of BFGS A. Rodomanov. [arXiv]	2024
Gradient Methods for Stochastic Optimization in Relative Scale Y. Nesterov and A. Rodomanov. [arXiv]	2023

Conference papers

Non-Convex Federated Optimization under Cost-Aware Client Selection X. Jiang, A. Rodomanov, S. Stich. ICLR 2026. [arXiv]	2026
Composite Optimization with Error Feedback: the Dual Averaging Approach Y. Gao, A. Rodomanov, J. Rack, S. Stich. ICLR 2026. [arXiv]	2026
Accelerated Distributed Optimization with Compression and Error Feedback Y. Gao, A. Rodomanov, J. Rack, S. Stich. AISTATS 2026. [arXiv]	2026
DADA: Dual Averaging with Distance Adaptation M. Moshtaghifar, A. Rodomanov, D. Vankov, S. Stich. ICLR 2026. [arXiv]	2026
Exploiting Similarity for Computation and Communication-Efficient Decentralized Optimization Y. Takezawa, X. Jiang, A. Rodomanov, S. Stich. ICML 2025:58359-58405. [url] [pdf] [arXiv]	2025
Optimizing (L_0, L_1)-Smooth Functions by Gradient Methods D. Vankov, A. Rodomanov, A. Nedich, L. Sankar, S. Stich. ICLR 2025. [url] [pdf] [arXiv]	2025
Stabilized Proximal-Point Methods for Federated Optimization X. Jiang, A. Rodomanov, S. Stich. NeurIPS 2024:99735-99772. [url] [pdf] [arXiv]	2024
Universality of AdaGrad Stepsizes for Stochastic Optimization: Inexact Oracle, Acceleration and Variance Reduction A. Rodomanov, X. Jiang, S. Stich. NeurIPS 2024:26770-26813. [url] [pdf] [arXiv]	2024
Non-convex Stochastic Composite Optimization with Polyak Momentum Y. Gao, A. Rodomanov, S. Stich. ICML 2024:14826-14843. [url] [pdf] [arXiv]	2024
Federated Optimization with Doubly Regularized Drift Correction X. Jiang, A. Rodomanov, S. Stich. ICML 2024:21912-21945. [url] [pdf] [arXiv]	2024
Universal Gradient Methods for Stochastic Convex Optimization A. Rodomanov, A. Kavis, Y. Wu, K. Antonakopoulos, V. Cevher. ICML 2024:42620-42646. [url] [pdf] [arXiv]	2024
Polynomial Preconditioning for Gradient Methods N. Doikov and A. Rodomanov. ICML 2023:8162-8187. [url] [pdf] [arXiv]	2023
A Superlinearly-Convergent Proximal Newton-Type Method for the Optimization of Finite Sums A. Rodomanov and D. Kropotov. ICML 2016:2597-2605. [url] [pdf] [supplementary] [code]	2016
Primal-Dual Method for Searching Equilibrium in Hierarchical Congestion Population Games P. Dvurechensky, A. Gasnikov, E. Gasnikova, S. Matsievsky, A. Rodomanov, I. Usik. DOOR-SUP 2016:584-595. [url] [arXiv]	2016
Putting MRFs on a Tensor Train A. Novikov, A. Rodomanov, A. Osokin, D. Vetrov. ICML 2014:811-819. [url] [pdf] [supplementary] [poster] [slides] [code]	2014

Journal articles

Subgradient ellipsoid method for nonsmooth convex problems A. Rodomanov and Y. Nesterov. Math. Program. [url] [arXiv]	2022
New Results on Superlinear Convergence of Classical Quasi-Newton Methods A. Rodomanov and Y. Nesterov. J. Optim. Theory Appl. 188:744-769. [url] [arXiv]	2021
Rates of superlinear convergence for classical quasi-Newton methods A. Rodomanov and Y. Nesterov. Math. Program. [url] [arXiv]	2021
Greedy Quasi-Newton Methods with Explicit Superlinear Convergence A. Rodomanov and Y. Nesterov. SIAM J. Optim. 31(1):785-811. [url] [arXiv]	2021
Smoothness Parameter of Power of Euclidean Norm A. Rodomanov and Y. Nesterov. J. Optim. Theory Appl. 185:303-326. [url]	2020
A Randomized Coordinate Descent Method with Volume Sampling A. Rodomanov and D. Kropotov. SIAM J. Optim. 30(3):1878-1904. [url] [arXiv]	2020

Workshop papers

DADA: Dual Averaging with Distance Adaptation

2024

M. Moshtaghiar, A. Rodomanov, D. Vankov, S. Stich. OPT2024 Workshop @ NeurIPS. [\[url\]](#) [\[pdf\]](#)

A Newton-type Incremental Method with a Superlinear Rate of Convergence

2015

A. Rodomanov and D. Kropotov. OPT15@NIPS. [\[url\]](#) [\[pdf\]](#)

TALKS AT CONFERENCES AND SEMINARS

Efficient Gradient Methods for Functions with Super-Quadratic Curvature

Jun 2026

Optimization Seminar at UCLouvain [\[slides\]](#)

Louvain-la-Neuve, Belgium

VOCAL Optimization Conference [\[slides\]](#)

Mosonmagyaróvár, Hungary

Efficient Distance-Adaptive Subgradient Methods

Aug 2025

Optimization Seminar at UCLouvain [\[slides\]](#)

Louvain-la-Neuve, Belgium

Optimizing (L_0, L_1) -Smooth Functions by Gradient Methods

Nov, Dec 2024

Research Seminar at Université Grenoble Alpes [\[slides\]](#)

Grenoble, France

Research Seminar at Weierstrass Institute [\[slides\]](#)

Berlin, Germany

Adaptive Gradient Methods for Stochastic Optimization

Oct 2024

Blue Yonder Series on Optimization for Machine Learning [\[slides\]](#)

online

Universality of AdaGrad Stepsizes for Stochastic Optimization: Inexact Oracle, Acceleration and Variance Reduction

Jun, Jul, Aug 2024

FGS Conference on Optimization [\[slides\]](#)

Gijón, Spain

EURO Conference on Operational Research [\[slides\]](#)

Copenhagen, Denmark

ALGOPT Workshop on Algorithmic Optimization [\[slides\]](#)

Louvain-la-Neuve, Belgium

Universal Gradient Methods for Stochastic Convex Optimization

Mar, Apr 2024

MOP Research Seminar on Mathematical Optimization [\[slides\]](#)

online

Research Seminar at CORE [\[slides\]](#)

Louvain-la-Neuve, Belgium

Gradient Methods for Stochastic Optimization in Relative Scale

Mar, May 2023

Research Seminar of DAO team at Université Grenoble Alpes [\[slides\]](#)

Grenoble, France

SIAM Conference on Optimization (OP23) [\[slides\]](#)

Seattle, USA

Modern analysis of local convergence for classical quasi-Newton methods

Mar 2023

Maths Job Market Seminar at Toulouse School of Economics [\[slides\]](#)

Toulouse, France

Universal Stochastic Gradient Methods for Convex Optimization

Jan 2023

Research Seminar at CISPA Helmholtz Center for Information Security [\[slides\]](#)

Saarbrücken, Germany

Subgradient Ellipsoid Method for Nonsmooth Convex Problems

May 2022

20th French-German-Portuguese Conference on Optimization (FGP22) [\[slides\]](#)

Porto, Portugal

New Results on Superlinear Convergence of Classical Quasi-Newton Methods

Mar, Jul 2021

XIII Symposium of Numerical Analysis and Optimization [\[slides\]](#)

Curitiba, Brazil (online)

18th Workshop on Advances in Continuous Optimization (EUROPT 2021) [\[slides\]](#)

Toulouse, France (online)

Greedy Quasi-Newton Method with Explicit Superlinear Convergence

Jun, Aug, Sep, Oct 2019

17th Workshop on Advances in Continuous Optimization (EUROPT 2019) [\[slides\]](#)

Glasgow, UK

Sixth International Conference on Continuous Optimization (ICCOPT 2019) [\[slides\]](#)

Berlin, Germany

19th French-German-Swiss Conference on Optimization (FGS'2019) [\[slides\]](#)

Nice, France

Seminar in Mathematical Engineering at UCLouvain [\[slides\]](#)

Louvain-la-Neuve, Belgium

Lecture: Introduction to Stochastic Optimization

Aug 2018

DeepBayes Summer School [\[slides\]](#) [\[video\]](#)

Moscow, Russia

Adaptive gradient methods for stochastic and online optimization

Feb 2018

Seminar on Bayesian Methods in Machine Learning [\[slides\]](#)

Moscow, Russia

Incremental Newton Method for Big Sums of Functions

Oct 2016

Seminar on Stochastic Analysis in Problems, IUM [\[slides \(in Russian\)\]](#) [\[video \(in Russian\)\]](#)

Moscow, Russia

A Superlinearly-Convergent Proximal Newton-Type Method for the Optimization of Finite Sums International Conference on Machine Learning (ICML) [slides] [video]	Jun 2016 New York, USA
Optimization Methods for Big Sums of Functions Deep Machine Intelligence Workshop at Skoltech [slides]	Jun 2016 Moscow, Russia
Incremental Newton Method for Minimizing Big Sums of Functions HSE off-site seminar on Machine Learning [slides]	May 2016 Voronovo, Russia
Introduction to the Tensor Train Decomposition and Its Applications in Machine Learning Seminar on Applied Linear Algebra at HSE [slides]	Mar 2016 Moscow, Russia
Proximal Incremental Newton Method Seminar on Bayesian Methods in Machine Learning [slides]	Feb 2016 Moscow, Russia
Probabilistic Graphical Models: a Tensorial Perspective International Conference on Matrix Methods in Mathematics and Applications (MMMA) [slides]	Aug 2015 Moscow, Russia
A Fast Incremental Optimization Method with a Superlinear Rate of Convergence Summer School on Control, Information and Optimization [slides]	Jun 2015 Solnechnogorsk, Russia
Markov Chains and Spectral Theory Seminar on Bayesian Methods in Machine Learning [slides (in Russian)]	Oct 2014 Moscow, Russia
Low-Rank Representation of MRF Energy by means of the TT-Format SIAM Conference in Imaging Science (SIAM-IS) [slides]	May 2014 Hong-Kong, China
Fast Gradient Method Seminar on Bayesian Methods in Machine Learning [slides (in Russian)]	Apr 2014 Moscow, Russia
TT-Decomposition for Compact Representation of Tensors Seminar on Bayesian Methods in Machine Learning [slides (in Russian)]	Oct 2013 Moscow, Russia

POSTERS

Universality of AdaGrad Stepsizes for Stochastic Optimization: Inexact Oracle, Acceleration and Variance Reduction Joint with X. Jiang and S. Stich. NeurIPS 2024. [pdf]	Dec 2024 Vancouver, Canada
Universal Gradient Methods for Stochastic Convex Optimization Joint with A. Kavis, Y. Wu, K. Antonakopoulos, V. Cevher. ICML 2024. [pdf]	Jul 2024 Vienna, Austria
Randomized Minimization of Eigenvalue Functions Joint with Y. Nesterov. Optimization and Statistical Learning Workshop. [pdf]	Jan 2023 Les Houches, France
Quasi-Newton and Second-Order Methods for Convex Optimization Joint with N. Doikov and Y. Nesterov. ICTEAM Welcome Day. [pdf]	Oct 2021 Louvain-la-Neuve, Belgium
A Superlinearly-Convergent Proximal Newton-Type Method for the Optimization of Finite Sums Joint with D. Kropotov. ICML 2016. [pdf]	Jun 2016 New York, USA
A Newton-type Incremental Method with a Superlinear Convergence Rate Joint with D. Kropotov. OPT15@NIPS. [pdf]	Dec 2015 Montreal, Canada
A Fast Incremental Optimization Method with a Superlinear Rate of Convergence Joint with D. Kropotov. Microsoft Research PhD Summer School. [pdf]	Jul 2015 Cambridge, UK
Putting MRFs on a Tensor Train Joint with A. Novikov, A. Osokin and D. Vetrov. ICML 2014. [pdf]	Jun 2014 Beijing, China

RESEARCH VISITS

UCLouvain Hosted by Yurii Nesterov .	Jun 2026 Louvain-la-Neuve, Belgium
UCLouvain Hosted by Geovani Grapiglia .	Aug 2025 Louvain-la-Neuve, Belgium
Weierstrass Institute Hosted by Vladimir Spokoyny .	Dec 2024 Berlin, Germany
Université Grenoble Alpes Hosted by Anatoli Juditsky .	Nov 2024 Grenoble, France
UCLouvain Hosted by Yurii Nesterov .	Apr 2024 Louvain-la-Neuve, Belgium
DAO team at Université Grenoble Alpes Hosted by Jérôme Malick .	Mar 2023 Grenoble, France
CISPA Helmholtz Center for Information Security Hosted by Sebastian Stich .	Jan 2023 Saarbrücken, Germany
Laboratory for Information and Inference Systems (LIONS) at EPFL Hosted by Volkan Cevher .	Jul, Nov 2022 Lausanne, Switzerland

AWARDS

Increased State Academic Scholarship for research and academic achievements, at Higher School of Economics	2017
Golden HSE Award in the Silver Nestling nomination, at Higher School of Economics	2016
Scholarship of the Lukoil Fund , at Higher School of Economics	2016
Ilya Segalovich Scholarship (from Yandex), at Higher School of Economics	2016
Travel award , at International Conference on Machine Learning (ICML)	2016
Best thesis award (1st place), at Lomonosov Moscow State University	2015

TEACHING EXPERIENCE

Lectures on Modern Optimization Methods , lectures Graduate-level course at Saarland University . Joint with Sebastian Stich .	2025 Saarbrücken, Germany
Optimization Models and Methods II , exercise sessions Graduate-level course at UCLouvain . Lectures by François Glineur and Geovani Grapiglia .	2021–22 Louvain-la-Neuve, Belgium
Optimization Methods in Machine Learning , exercise sessions Graduate-level course at Lomonosov Moscow State University , Yandex School of Data Analysis and Moscow Institute of Physics and Technology . Lectures by Dmitry Kropotov .	2015–18 Moscow, Russia
Continuous Optimization , exercise sessions Undergraduate-level course at Higher School of Economics . Lectures by Dmitry Kropotov .	2017–18 Moscow, Russia
Machine Learning , exercise sessions Graduate-level course at Skoltech . Lectures by Victor Kitov .	2015 Moscow, Russia

REVIEWING

- **Journals:** [Mathematical Programming](#), [SIAM Journal on Optimization \(SIOPT\)](#), [Journal of Optimization Theory and Applications \(JOTA\)](#), [Journal of Machine Learning Research \(JMLR\)](#), [Optimization Methods and Software](#), [Applied Mathematics & Optimization \(AMOP\)](#), [Computational Optimization and Applications \(COAP\)](#), [Automatica](#).
- **Conferences:** [Conference on Neural Information Processing Systems \(NeurIPS\)](#), [International Conference on Machine Learning \(ICML\)](#), [ACM Symposium on Theory of Computing \(STOC\)](#).

TECHNICAL SKILLS

- **Programming languages:** Python, C++, Rust.
- **Machine learning & Scientific computing:** NumPy, SciPy, PyTorch, Pandas, Scikit-learn, Eigen.
- **Development & Code quality:** Ruff, uv, MyPy, pytest, CMake, Clang-Format, Clang-Tidy, Sanitizers.
- **Version control & CI/CD:** Git, GitHub, GitHub Actions, Pre-commit hooks.
- **Document preparation:** LaTeX.